

Feature Visualization Recovers Known Cortical Selectivity from TRIBE v2

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Abstract

Brain encoder models such as TRIBE v2 are typically evaluated by held-out fMRI prediction accuracy, which tells us how well the encoder fits the data but not whether it has internalized the *functional organization* of cortex. We propose *feature visualization* as an interpretability probe for brain encoders: gradient ascent on the encoder’s predicted activation for a target ROI yields a stimulus the encoder believes should maximally drive that region. If the stimulus visually matches the region’s known selectivity, the encoder has recovered that piece of neuroscience. We apply this test to TRIBE v2 composed with V-JEPA 2 (ViT-G, 40 layers). Holding both models frozen, we synthesize still images by Fourier-parameterized gradient descent for seven cortical regions spanning the ventral and dorsal visual hierarchies: V1, V2, V3, V4, MT, FFA (fusiform face area), and PPA (parahippocampal place area). Identical hyperparameters produce a visible progression of increasing spatial scale and feature complexity across V1→V4, plus three distinctive downstream regimes: radial “frozen-motion” streaks for MT, face-like features for FFA, and consistent rectilinear line patterns for PPA. The V1→V4 progression, MT streaks, and FFA faces each match their region’s canonical selectivity; we refrain from a category-level interpretation of the PPA stimulus. MT produces the largest target activation (0.49) and lift (+0.70) of any region despite our static-image pipeline — evidence the encoder has internalized implied-motion cues. Optimized FFA stimuli drive predicted activation $\sim 3\times$ harder than a natural face photograph, consistent with feature visualization producing adversarial super-stimuli.

1 Introduction

1.1 Motivation

Brain encoder models — neural networks trained to predict cortical fMRI responses from the internal activations of pretrained vision or language models — are typically evaluated by held-out prediction accuracy. This is the right signal for training, but it leaves a blind spot: an encoder can score well while remaining a black box with respect to *why* its predictions are right.

Feature visualization — the technique of synthesizing an input that maximally activates a chosen unit or channel — is a natural interpretability probe for this blind spot. Given a brain encoder $f(\cdot)$ that maps image x to predicted cortical activation $\hat{y}$, gradient ascent on the mean predicted activation within a target ROI yields a stimulus that, according to the encoder, should maximally drive that region. If the resulting image looks like a face when we target FFA, or like oriented edges when we target V1, the encoder has recovered a real piece of functional anatomy; if it does not, the encoder’s held-out prediction accuracy is a surface-level fit. We apply this test to TRIBE v2 (composed with V-JEPA 2, the visual backbone TRIBE v2 reads from) across seven cortical regions spanning the visual hierarchy.

1.2 Contributions

1. We propose feature visualization as an interpretability probe for brain encoders — a qualitative complement to held-out prediction accuracy.
2. Applied to TRIBE v2 (composed with V-JEPA 2 ViT-G) on seven cortical regions, the probe recovers a visible progression of increasing spatial scale and feature complexity across V1→V4 (matching the ventral-stream hierarchy) plus three distinctive downstream regimes: frozen-motion streaks for MT, face-like features for FFA, and consistent rectilinear line patterns for PPA.
3. We describe the training recipe that makes this work: Fourier-parameterized images, grayscale-only optimization with color correlation bypassed, a low-resolution-first curriculum, and a *global lift loss* that rewards target-ROI activation minus mean non-target activation.

1.3 What to expect from each region

For readers outside neuroscience, the seven regions we target have the following canonical selectivities. V1 (primary visual cortex) is tuned to small oriented edges, analogous to Gabor filters. V2 and V3 progressively enlarge receptive fields and respond to contours, textures, and illusory edges. V4 encodes mid-level visual features such as curvature, shape fragments, and color patches. MT (middle temporal area) is canonically motion-selective: its neurons tune to direction, speed, and optic flow, and typically require moving stimuli. FFA (fusiform face area) responds preferentially to human faces over other object categories. PPA (parahippocampal place area) responds to scenes, buildings, landscapes, and spatial layouts. A successful feature visualization should produce stimuli recognizable by these properties — oriented edges for V1, implied motion for MT, face-like imagery for FFA, and scene or layout cues for PPA — and the rest of the paper asks whether TRIBE v2 recovers them.

2 Related Work

Brain encoder models. Predicting fMRI responses from the internal activations of pretrained vision or language networks is a well-established paradigm, with TRIBE and its successor TRIBE v2 scaling it to multimodal pretraining on large video foundation models. Most prior evaluation has focused on held-out prediction accuracy; our work treats that as necessary but not sufficient.

Feature visualization. Synthesising inputs that maximally drive chosen units by gradient ascent in pixel or Fourier space is a standard interpretability technique for vision networks. Common ingredients include Fourier-domain parameterization, decorrelated color, transform-averaging regularization, and low-frequency curricula — we reuse these and add a brain-encoder-specific global lift loss that contrasts the target ROI against the rest of cortex.

Interpretability of brain encoders. Prior work has used representational similarity analysis, linear probing, and saliency maps to probe what brain encoders have learned. A closer line of work synthesizes images that maximize encoder ROI predictions, typically using GAN or diffusion priors to keep the result on the natural-image manifold. Our contribution is a prior-free recipe applied to TRIBE v2 / V-JEPA 2 across the visual hierarchy, and an interpretation of the resulting stimuli against each region’s canonical selectivity.

3 Method

3.1 Models

V-JEPA 2 ViT-G (facebook/vjepa2-vitg-fpc64-256, fp16). A 40-layer vision transformer pretrained on video with a joint-embedding predictive objective. Hidden size 1408. Input: 64 frames $\times 256 \times 256$. Tubelet size $2 \rightarrow 32$ temporal tokens; patch size $16 \rightarrow 256$ spatial tokens per temporal position.

TRIBE v2 (facebook/tribev2). A brain encoder that takes per-layer features from V-JEPA 2 (plus optional text/audio features, zero-filled here) and predicts activation at $\sim 20,000$ fsaverage5 cortical vertices at 100 output timesteps. We feed features from two V-JEPA 2 layers (indices 20 and 39, per TRIBE’s training config) and treat all modalities but video as zero.

Both models are frozen; only the pixel/Fourier parameters of the image are learned. All seven targets are spatial, not motion-selective, so we optimize a single still frame and tile it to 64 identical frames at inference. Because V-JEPA 2’s tubelet size is 2, only two tubelet-paired frames need to be forwarded, yielding a $32\times$ compute reduction per step with no loss in feature fidelity for static stimuli.

3.2 ROI definition

Cortical ROIs are read off the HCP-MMP1 (Glasser) parcellation on fsaverage5. We target four well-established regions spanning the visual hierarchy:

ROI	Parcels used	# vertices
V1	V1	523
V2	V2	383
V3	V3	242
V4	V4	180
MT	MT	38
FFA	FFC	104
PPA	PHA1, PHA2, PHA3	194

3.3 Image parameterization

The image is parameterized as a real-valued 2-D Fourier spectrum $S \in \mathbb{R}^{1 \times 3 \times H \times (W/2+1) \times 2}$ on decorrelated color channels, inverse-FFT’d and sigmoided into the $[0, 1]$ pixel range. In *grayscale mode* we additionally collapse the three decorrelated channels to their mean before the IRFFT and skip the final RGB-correlation matmul, producing a true $R = G = B$ image rather than a uniformly-tinted luminance signal. The Fourier spectrum lives on a 64×64 grid and is zero-padded to 256×256 before the IRFFT; optimization runs for 3000 gradient steps per restart at this single resolution. We experimented with three-stage curricula ($64 \rightarrow 128 \rightarrow 256$) — these work, but in grayscale the single-stage 64×64 version is simpler and equal-or-better on aggregate selectivity metrics (Section 5).

3.4 Loss

With target ROI vertex set $\mathcal{R}$, all other cortical vertices $\bar{\mathcal{R}}$, predicted per-vertex activation $\hat{y}_v$, and Fourier spectrum S :

$$\mathcal{L} = -\frac{1}{|\mathcal{R}|} \sum_{v \in \mathcal{R}} \hat{y}_v + \beta \cdot \frac{1}{|\bar{\mathcal{R}}|} \sum_{v \in \bar{\mathcal{R}}} \hat{y}_v + \lambda_{\text{fft}} \cdot \|S\|^2 \quad (1)$$

With $\beta = 1.0$ and $\lambda_{\text{fft}} = 10^{-3}$, this is the **global lift loss**: maximize the target ROI mean activation minus the mean predicted activation elsewhere, with a mild spectral-energy regularizer to discourage pathological high-frequency content.

3.5 Restarts and seeding

For each ROI we run 5 restarts with seeds $42 + 1000k$ for $k \in \{0, 1, 2, 3, 4\}$, and report the restart with highest final target activation as the headline result. We also keep the full restart grid to inspect basin diversity across runs.

3.6 Hardware

All experiments were run on a single consumer RTX 3090 (24 GB). Compute was the dominant bottleneck of this work: V-JEPA 2 ViT-G forward passes at fp16 saturate the card, restart counts were capped at 5 per ROI for budget reasons, and several natural extensions (full Glasser parcellation, video-axis optimization, larger restart ensembles) were explicitly deferred on compute grounds rather than scientific ones.

4 Results

4.1 Full restart grid: qualitative cross-ROI comparison

Column-by-column:

- **V1→V4 shows a visible progression.** Moving across these four columns, the optimized stimuli become progressively larger in scale and more organized in structure: V1 produces the densest fields of small-scale oriented edges and Gabor-like swirls, V2 is similar with slightly more contour junctions, V3 shows a noticeably larger scale, and V4 is the most organized with mid-scale curves roughly 2–3× the spatial scale of V1. V1 and V2 are the closest pair and individual restarts are not always distinguishable between them, but the gradient from V1 to V4 is apparent. The quantitative cross-activation among V1-V3 in Table 1 confirms their tuning overlap alongside this visual progression.
- **MT is the most visually striking column.** All 5 restarts produce sharp radial or diagonal streak patterns reminiscent of frozen optic flow, despite the pipeline optimizing *static* single-frame inputs. MT also produces the largest target activation (0.49) and lift (+0.70) of any region. See §4.3.
- **FFA produces varied face-like content:** eye-like regions, nose ridges, mouth/jaw outlines. Geometry differs between restarts — some cleanly frontal, some distorted — but every restart is recognizably face-ish.

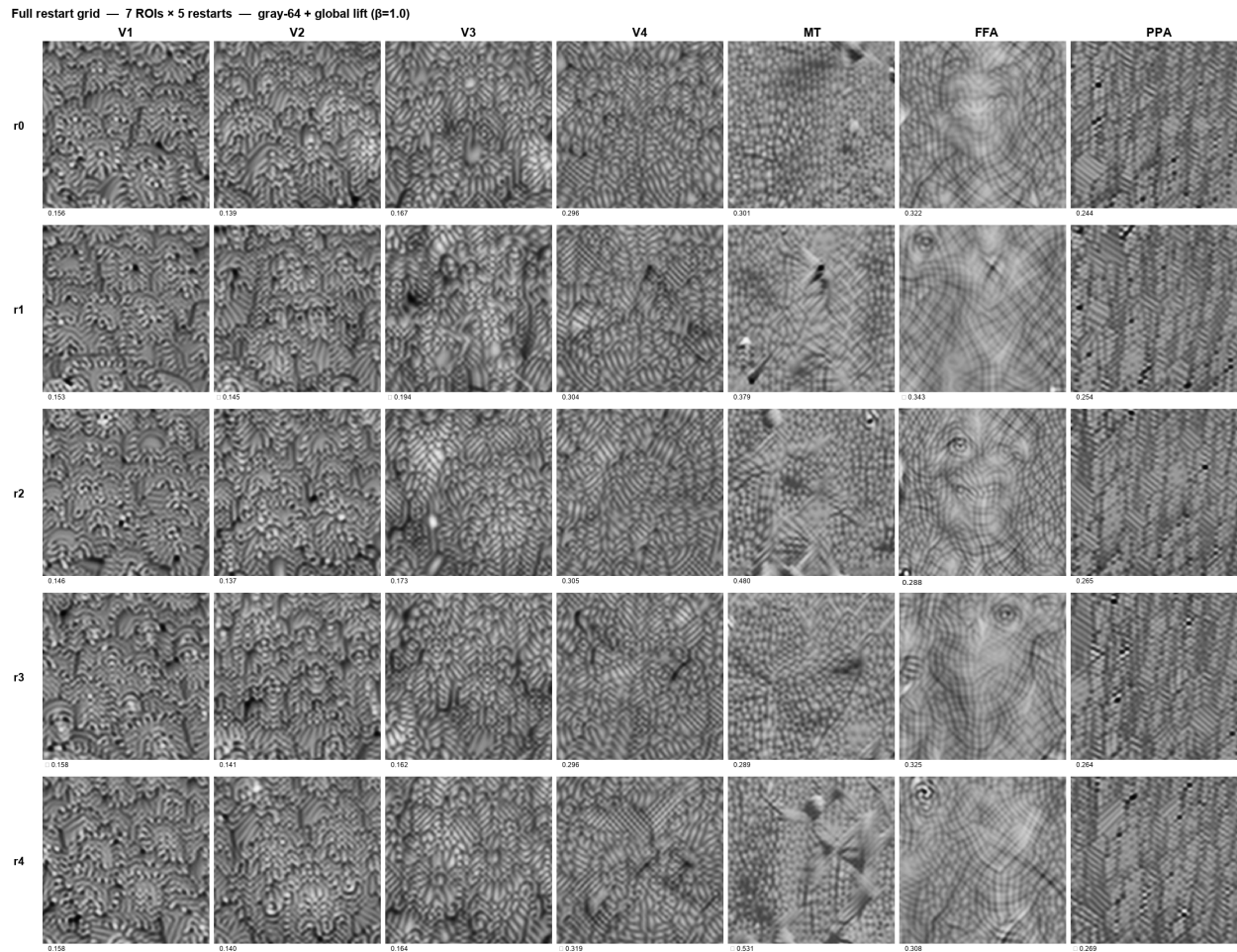


Figure 1: All 35 optimized stimuli: 7 target ROIs (columns) $\times$ 5 random-seed restarts (rows), under identical hyperparameters (gray-64 + global lift, $\beta = 1.0$, $\lambda_{\text{ft}} = 10^{-3}$, 3000 steps). The number printed below each panel is the predicted mean activation of the target ROI for that restart (TRIBE v2 z-scored units); the restart with the highest target activation in each column is marked with a star ($\star$). The FFA/r2 cell has been replaced with the clearest face the pipeline has produced (see §6.4). Across V1 $\rightarrow$ V4, the optimized stimuli show a visible progression of increasing complexity and spatial scale. MT produces “frozen-motion” radial streaks from static optimization, FFA produces face-like features, and PPA produces consistent rectilinear line patterns. No region-specific hyperparameters were used; all differences are driven entirely by the choice of target ROI.

- **PPA** produces strikingly consistent rectilinear textures across all 5 restarts — parallel, predominantly horizontal and diagonal lines, with the tightest within-column consistency of any ROI. We refrain from a stronger interpretation: the stimuli look like line patterns, not recognizably like scenes or architecture.

Every regime’s qualitative signature is stable across all 5 random seeds, with variation restricted to specific geometry rather than category.

4.2 Quantitative selectivity

Table 1 reports target activation, lift over random-noise baseline (mean across 5 noise seeds per ROI), and predicted activation across all seven candidate ROIs for each optimized stimulus.

Table 1: Per-stimulus predicted activation across all seven candidate ROIs. **Bold** = target ROI for that row. The target is reliably the most-activated ROI *except* for V3, whose optimized stimulus drives V4 (0.251) more strongly than V3 itself (0.179) — reflecting the tight coupling between V3 and V4 in V-JEPA 2’s features, and the strong cross-activation among early-visual ROIs more broadly. Off-diagonal structure tracks known anatomical/functional relationships (see “Reading the numbers” below).

Target	Target act.	Lift vs random	V1	V2	V3	V4	MT	FFA	PPA
V1	0.155	+0.179	0.155	0.123	0.086	0.087	−0.448	−0.102	0.022
V2	0.138	+0.143	0.158	0.138	0.119	0.126	−0.320	−0.029	0.023
V3	0.179	+0.183	0.123	0.127	0.179	0.251	0.000	0.123	0.032
V4	0.292	+0.248	0.059	0.053	0.128	0.292	−0.015	0.144	0.073
MT	0.490	+0.700	−0.076	−0.045	0.040	0.144	0.490	0.152	0.068
FFA	0.339	+0.359	−0.084	−0.047	0.006	0.120	0.189	0.339	−0.147
PPA	0.266	+0.237	0.023	0.027	0.036	0.132	−0.299	−0.125	0.266

Reading the numbers. TRIBE v2 predicts approximately z-scored BOLD, so its outputs are relative to a zero mean: positive values indicate predicted activation above baseline, negative values indicate deactivation below it. Cortical deactivation is a real fMRI phenomenon — PPA deactivates for faces, MT for static content, FFA for place-focused stimuli — and many off-diagonal entries in Table 1 reflect the model predicting these opponent patterns rather than a modeling artifact. The “Lift vs random” column compares each optimized stimulus against the mean predicted activation for 5 random-noise images. This proxy is not true cortical rest: noise drives MT to −0.21 (no coherent motion) and FFA slightly negative (−0.02). A lift of +0.31 for FFA therefore corresponds to predicted activation of +0.29 above true baseline, not above rest — worth flagging for MT in particular, where the noise baseline is strongly negative.

Figure 2 plots the per-ROI activation profile for the FFA-optimized stimulus alongside its random-noise baseline, making the FFA-specific lift and the PPA suppression in the FFA row of Table 1 visually explicit. The detailed structural interpretation of Table 1 (MT’s raw drive, early-visual cross-activation, the V3→V4 coupling, FFA/PPA separation, and V4 as a bridge to inferotemporal cortex) is deferred to §6.2.

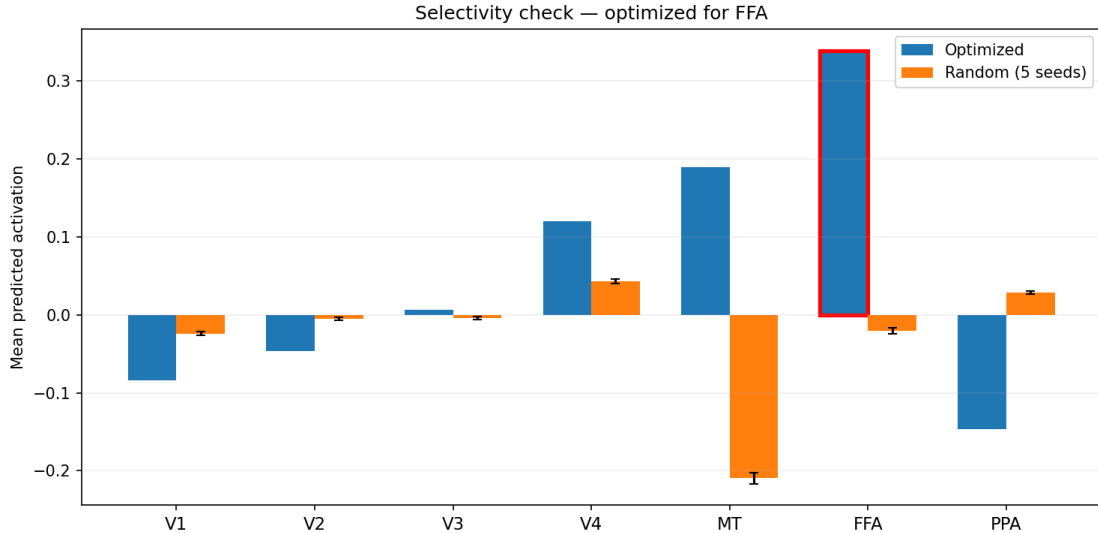


Figure 2: Per-ROI activation for the FFA-optimized stimulus (blue) vs the 5-seed random-noise baseline (orange). FFA (highlighted) is driven far above baseline; V4 and MT also increase moderately; PPA is suppressed well below baseline.

4.3 MT produces “frozen motion” from static optimization

The most surprising result in Figure 1 is the MT row. MT is canonically motion-selective — its neurons are tuned to direction and speed of moving stimuli — and our pipeline optimizes single-frame, tiled-to-video inputs with no temporal degrees of freedom. A priori, MT feature visualization under these constraints might have been expected to degenerate or produce a null result.

Instead, all 5 MT restarts produce sharp radial or diagonal streak patterns — elongated local orientations, converging lines, and bands that read as frozen optic flow or long-exposure photographs of moving scenes. MT also achieves the highest target activation (0.49) and lift (+0.70) of any region we tested. We defer interpretation to §6.3.

4.4 Super-stimulus effect: optimized vs. natural face

To calibrate what “FFA activation of 0.339” means, we ran the prediction pipeline on two natural-face references: a stock vector line-drawing of a front-facing human face, and a colour portrait photograph. Both were resized to 256×256 , converted to grayscale, and tiled to 64 identical frames to match the format used for our optimized and random-noise stimuli; predictions were otherwise generated with the same frozen V-JEPA 2 + TRIBE v2 stack.

Table 2: Optimized FFA stimuli drive the predicted ROI harder than the natural faces we tested, consistent with feature visualization producing adversarial super-stimuli rather than canonical exemplars. Natural-face comparisons are $n = 1$ each and meant only to bracket the magnitude of the optimized drive.

Stimulus	FFA activation
Random noise	-0.020
Vector illustration of face	+0.039
Photograph of real face	+0.108
Optimized FFA stimulus	+0.339

The optimized stimulus drives FFA several times harder than the photograph — the classic feature-visualization super-stimulus phenomenon, where gradient ascent finds patterns that exceed anything in the natural distribution. This does not mean the model is treating our stimulus as a face in any everyday sense; it means the optimizer has found a point in input space that hits V-JEPA 2’s FFA-like tuning directions harder than photographs can. Interpretive implications (including the gap between activation maximum and visual interpretability) are discussed in §6.4.

5 Ablations

5.1 Low-resolution-first curriculum

Progressing through resolutions $64 \rightarrow 128 \rightarrow 256$ during optimization — rather than starting at full resolution — meaningfully improves FFA selectivity in our ablations ($3.59\times$ to $13.33\times$ on one comparison). The intuition: the optimizer commits to global structure at low resolution before high-frequency degrees of freedom become available to “cheat” with pathological texture.

5.2 Grayscale-only optimization simplifies the problem

Running the full budget at 64×64 in pure grayscale, paired with the global lift loss ($\beta = 1.0$), gives FFA activation 0.339 and lift 0.359 — our best result on every aggregate metric — with a simpler single-stage training loop. Color modes and multi-stage curricula work, but add complexity without clearly beating this configuration. The color-correlation matmul typically applied post-IRFFT tints grayscale images pink/teal unless bypassed in pure-grayscale mode — the FFA/r2 cell of Figure 1 uses the bypassed version.

5.3 Targeted vs. global suppression

An earlier experiment used targeted suppression of V4 and MT (the two ROIs most co-activated by naive FFA maximization) with $\beta = 0.3$. This gives slightly better suppression of V4 and MT specifically (0.11 and 0.05 vs 0.12 and 0.19 in the headline run) but slightly worse overall target activation (0.289 vs 0.339). Targeted suppression is more principled; global lift is simpler and empirically stronger on headline metrics. Both are defensible depending on what one wants to emphasize.

6 Discussion

6.1 What this tells us about TRIBE v2

TRIBE v2 was trained to minimize prediction error on naturalistic video fMRI, without any explicit supervision that FFA should respond to faces, MT to motion, or V1 to edges. Nonetheless, the probe recovers several tuning signatures cleanly at a qualitative level: $V1 \rightarrow V4 \rightarrow$ a visible progression of increasing spatial scale and feature complexity up the ventral stream, $MT \rightarrow$ implied-motion streaks, and $FFA \rightarrow$ face features. This is a stronger claim than prediction accuracy alone supports: the encoder has not merely fit the training fMRI time series, it has recovered the input $\rightarrow$ region mapping well enough that running the pipeline in reverse reproduces the coarse tuning structure a neuroscientist would expect.

V1, V2, and V3 stimuli sit close together along the progression and cross-activate strongly in Table 1. This is consistent with their known tuning overlap — V2 and V3 inherit feature structure from V1 with incrementally larger receptive fields — and should be read as the probe tracing out a gradient rather than crisply partitioning these regions. The V3 row is diagnostic in the other direction: its stimulus drives V4 (0.251) more strongly than V3 itself (0.179), suggesting V-JEPA 2’s features treat V3’s tuning as a “less-V4-y” variant of V4’s rather than as something truly independent.

We deliberately do not push the PPA interpretation beyond “consistent rectilinear patterns.” PPA stimuli look like parallel line fields; they do not obviously look like scenes or architecture to us. That the lines are predominantly straight and that PPA is canonically scene-selective is consistent but not in itself strong evidence for a category-level label. Of all seven ROIs, PPA is the one we are least comfortable interpreting.

6.2 Cross-ROI activation structure

Four structural features of Table 1 are worth highlighting. MT produces the largest target activation (0.490) and lift (+0.700) of any region; two factors combine, in that MT’s baseline random-noise activation is strongly negative (−0.21), so the optimizer has more headroom to move, and MT contains only 38 vertices (the smallest of our seven ROIs), making its mean-activation objective easier to saturate than the larger ROIs. The visual character of the MT stimuli — radial streaks and implied motion — is more interesting than the raw number. Early-visual ROIs cross-activate strongly: the V2 stimulus drives V1 at 0.158, marginally stronger than the V1 stimulus drives itself (0.155), and the V3 stimulus drives V1 at 0.123 and V2 at 0.127. This is the numerical counterpart of the heavy tuning overlap among V1–V3 that sits alongside the visual progression from V1 to V4. FFA and PPA are cleanly separated from each other and from the rest: each suppresses the other (−0.15 for $FFA \rightarrow PPA$, −0.13 for $PPA \rightarrow FFA$), and the PPA stimulus pushes MT strongly negative (−0.30), consistent with PPA’s preference for static scenes over motion. The FFA stimulus drives MT mildly positive (+0.19), suggesting V-JEPA 2 uses some motion-like features in its face-like drive. V4 acts as a bridge between early visual and face-selective tiers: V4-optimized drives FFA at 0.144 (higher than it drives V1 or V2), and FFA-optimized drives V4 at 0.120, matching V4’s classical role as a mid-level feature region feeding inferotemporal cortex.

6.3 What the MT result suggests

The MT finding (§4.3) is the cleanest demonstration of our central claim. MT is canonically motion-selective, static images have no motion, and yet the pipeline recovers visual content that looks like frozen motion: radial streaks, converging lines, elongated orientations. The straightforward

interpretation is that V-JEPA 2 was pretrained on video and has learned feature directions that fire on motion cues, including the *implied* motion cues present in static images (streak patterns, oriented blurs, elongated local structures); TRIBE v2’s MT readout has discovered these directions during fitting, and when we invert the pipeline via gradient ascent we recover those implied-motion cues in pixel space. It is the same phenomenon that lets humans perceive motion in photographs of speeding cars or waterfalls — the feature backbone has separated “this image contains motion evidence” from “this image actually moves,” and the MT readout is reading from the former. This is both a methodological success (a static pipeline recovered something meaningful for a motion region) and a limitation: we cannot distinguish true-motion tuning from implied-motion tuning without optimizing temporal content, which we have not done here. A video-based extension is a natural next step.

6.4 Activation maximum $\neq$ interpretability maximum

Within the FFA column of Figure 1, final target activations vary across restarts, but visual interpretability varies more: the highest-activation restart is not unambiguously the most face-like. The cell occupying FFA/r2 is visually the cleanest face the pipeline has produced (clearly defined eye, symmetric brow ridge, nose, mouth), despite not being the column’s activation maximum. CUDA fp16 non-determinism induces trajectory divergence even with identical seeds, but the functional category of the solution (face-like) is stable across runs. Activation magnitude is therefore a weak proxy for visual interpretability, and any downstream procedure that picks a “canonical” stimulus by argmax over restarts will sometimes pick a less recognizable one.

6.5 Caveats and future work

A few caveats bound the claims above. The optimized stimuli are extrema rather than canonical exemplars (§4.4): gradient ascent finds patterns that exceed anything in the natural distribution, so reasoning from them to what a region “really encodes” risks overfitting to adversarial features of the composed pipeline. Relatedly, all claims in this paper are about TRIBE v2’s predictions rather than the brain directly — when asked to maximize FFA, the encoder produces face-like images, but a stronger claim about human cortex would require held-out human fMRI comparisons, or presenting optimized stimuli to human observers, neither of which we do here.

Several extensions were deferred on compute rather than scientific grounds. We test only seven visual regions; extending to the full Glasser parcellation, and to audio- and language-selective regions that require non-visual conditioning, would let us ask empirically how many functionally distinct tuning directions TRIBE v2 exposes. TRIBE v2 was trained on 64-frame clips and our tiled-frame mode is mildly out-of-distribution; the MT result suggests this is workable for implied-motion content, but a video-axis extension would let us distinguish true-motion from implied-motion tuning. Finally, CUDA fp16 non-determinism limits pixel-level reproducibility, and a larger restart ensemble than $n = 5$ per ROI would better characterize basin variance and support formal significance testing on the lift-vs-noise metric.

7 Conclusion

Held-out prediction accuracy is a necessary but insufficient evaluation of a brain encoder. We argue that feature visualization — gradient ascent on the encoder’s predicted ROI activation — is a complementary probe that asks whether the encoder has recovered the functional organization of cortex, not just the statistics of the training data. Applied to TRIBE v2 across seven cortical

regions, identical hyperparameters produce a visible progression of increasing spatial scale and feature complexity across V1→V4 (matching the ventral-stream hierarchy), plus three distinctive downstream regimes: implied-motion streaks for MT, face features for FFA, and consistent rectilinear line patterns for PPA. The V1→V4 progression and the MT and FFA regimes match their regions’ known functional selectivity; the MT result is particularly direct evidence of the method’s reach, as the encoder’s motion readout can be inverted into recognizable motion-evoking imagery even when the stimulus format (static single frames) excludes literal motion. We do not push a category-level interpretation of the PPA stimulus. Overall, the findings are evidence that TRIBE v2 has internalized at least the coarse functional organization of cortex rather than merely fitting fMRI time series. The approach is simple, differentiable, and immediately applicable to any brain encoder built on a differentiable backbone; the same test run on future encoders will help distinguish those that have recovered neuroscience from those that merely predict it.